

Majid Ghasemi

Ph.D. Student, Electrical & Computer Engineering, University of Waterloo

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RESEARCH INTERESTS

Social and multi-agent reinforcement learning, machine ethics, AI alignment. My work asks how ethical behaviour can be learned and transmitted between agents rather than hard-coded as a fixed reward.

EDUCATION

University of Waterloo

Waterloo, ON, Canada

Ph.D., Electrical & Computer Engineering

Sep 2025 – Present

Advisor: Mark Crowley

Focus: ethical decision making in multi-agent systems and social learning as a route to it

Wilfrid Laurier University

Waterloo, ON, Canada

M.A.Sc., Applied Computing (GPA A+)

Sep 2023 – May 2025

Advisor: Dariush Ebrahimi

Thesis: *Reinforcement Learning for Scheduling and Routing: Electric Vehicle Charging and Patrol Scheduling*

Imam Khomeini International University

Qazvin, Iran

B.Sc., Software Engineering (GPA A-)

Sep 2017 – Mar 2022

Advisor: Mahdi Bahaghighat

PUBLICATIONS

Name underlined. Reverse chronological within each tier.

Journal Articles

- [J1] **Real-Time Electric Vehicle Routing and Charging Optimization with Battery Life Consideration.**

Dariush Ebrahimi, Majid Ghasemi, Mohammad Kashefi, Fadi Alzhouri. *IEEE Internet of Things Journal*, 2025.

- [J2] **Citation Count Prediction Based on Google Scholar Profiles and Clarivate's Journal Citation Reports.**

Mahdi Bahaghighat, Leila Akbari, Majid Ghasemi, Qin Xin. *Information Research*, 2023.

- [J3] **A High-Accuracy Phishing Website Detection Method Based on Machine Learning.**

Mahdi Bahaghighat, Majid Ghasemi, Figen Ozen. *Journal of Information Security and Applications*, vol. 77, 2023.

Peer-Reviewed Conference Papers

- [C1] **Adaptive and AI-Driven Vehicle Patrol Scheduling with Integrated Emergency Response.**

Majid Ghasemi, Dariush Ebrahimi. *IECON 2025, 51st Annual Conference of the IEEE Industrial Electronics Society*, 2025.

- [C2] **Optimal and Heuristic Solutions for Efficient Vehicle Patrol Scheduling in Dynamic Urban**

Safety.

Majid Ghasemi, Ebrahim Sarkhouh, Fadi Alzhouri, Dariush Ebrahimi. *IEEE 34th International Symposium on Industrial Electronics (ISIE)*, 2025.

[C3] **Comparative Analysis of Reinforcement Learning for Reliable Real-Time EV Routing and Charging.**

Majid Ghasemi, Fadi Alzhouri, Dariush Ebrahimi. *21st International Conference on the Design of Reliable Communication Networks (DRCN)*, 2025.

[C4] **Navigating Smart Order Routing in Fixed Income Trading with Linear Programming and Dijkstra's Algorithm.**

Cassel Robson, Marko Misic, Erika Capper, Mila Cvetanovska, Majid Ghasemi, Fadi Alzhouri, Dariush Ebrahimi. *International Conference on the Dynamics of Information Systems*, 2024.

Peer-Reviewed Workshop Papers

[W1] **Learning When to Trust in Contextual Social Bandits.**

Majid Ghasemi, Mark Crowley. *19th European Workshop on Reinforcement Learning (EWRL)*, 2026.

[W2] **Rethinking AI Alignment: From Static Rewards to Social Reinforcement Learning.**

Majid Ghasemi, Mark Crowley. *Pluralistic Alignment Workshop, ICML*, 2026.

[W3] **Can Standard MARL Metrics Distinguish Communicative from Strategic Action?**

Majid Ghasemi, Mark Crowley. *PhilML Workshop, ICML*, 2026.

[W4] **Toward Virtuous Reinforcement Learning: A Critique and Roadmap.**

Majid Ghasemi, Mark Crowley. *Machine Ethics Workshop (From Formal Methods to Emergent Machine Ethics)*, AAAI, 2026.

Preprints and Technical Reports

[P1] **Comprehensive Survey of Reinforcement Learning: From Algorithms to Practical Challenges.**

Majid Ghasemi, Amirhossein Mousavi, Dariush Ebrahimi. *arXiv preprint*, 2024.

[P2] **Introduction to Reinforcement Learning.**

Majid Ghasemi, Dariush Ebrahimi. *arXiv preprint*, 2024.

HONOURS AND AWARDS

AIES Student Program Scholarship and Award	2026
Graduate Student Research and Development Award (×2), University of Waterloo	2026
International Doctoral Student Award, University of Waterloo	2025 – 2029
Academic Excellence Gold Medal, Wilfrid Laurier University	2025
Graduate Scholarship, Wilfrid Laurier University	2023 – 2025
Graduate Fellowship, Wilfrid Laurier University	2023 – 2025
Differential Tuition Fee Waiver, Wilfrid Laurier University	2023 – 2025
Excellence Students Tuition Fee Waiver, Imam Khomeini Intl. University	2017 – 2022

GRANTS

HU University of Applied Sciences Utrecht

Aug 2022 – Aug 2023

Research grant to study the Woman, Life, Freedom movement in Iran using computational social science and machine learning methods.

RESEARCH EXPERIENCE

UWECESML / CompThink Lab, University of Waterloo

Sep 2025 – Present

Graduate Researcher, advised by Mark Crowley

- Developing socially informed and ethically grounded methods for multi-agent reinforcement learning.
- Work spans virtue transmission in sequential social dilemmas, contextual bandits with strategic feedback, and constrained ethical agent design.

Edge Computing Lab, Wilfrid Laurier University

Sep 2023 – May 2025

Graduate Researcher, advised by Dariush Ebrahimi

- Reinforcement learning for intelligent transportation, focusing on EV fleet management, routing, and charging optimization.
- Developed exact and heuristic baselines for urban patrol scheduling under dynamic demand.

AIST Lab, Imam Khomeini International University

Jan 2021 – Mar 2022

Undergraduate Researcher, advised by Mahdi Bahaghighat

- Machine learning methods for phishing website detection.

INDUSTRY EXPERIENCE

Kisoji Biotechnology Inc.

Waterloo, ON, Canada

AI Research Intern, Machine Learning Department

May 2026 – Present

- Building the Expressibility Engine, a three-module system for antibody expression optimization.
- Domain-adaptive prediction with Bayesian project embeddings; RL-guided experimental design using GRPO with DPP-based diversity; sparse graph-based causal driver discovery.
- Reduces wet-lab iteration through active learning while producing interpretable guidance for antibody engineers.

INVITED TALKS AND PRESENTATIONS

Designing Virtuous Agents with Reinforcement Learning, Formal Ethics 2026, Buffalo, USA

Jul 2026

Optimal and Heuristic Solutions for Vehicle Patrol Scheduling, IEEE ISIE 2025

Jun 2025

TEACHING EXPERIENCE

Teaching Assistant

2021 – Present

- **University of Waterloo:** ECE108 Discrete Mathematics and Logic (S26), ECE406 Algorithm Design and Analysis (W26), ECE250 Data Structures and Algorithms, tutor (F25).
- **Wilfrid Laurier University:** CP468 Artificial Intelligence (F23, F24, W25), CP421 Data Mining (W24), CP312 Algorithm Design and Analysis (W24).
- **Imam Khomeini Intl. University:** Artificial Intelligence, Data Mining, Fundamentals of Programming (2021 – 2022).
- Guest lectured selected topics, authored quizzes and project specifications, designed and evaluated term

projects, and held regular office hours.

ACADEMIC SERVICE

Program Committee

- AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2026

Reviewer

- Reinforcement Learning in Big Worlds Workshop, Reinforcement Learning Conference (RLC), 2026
- PhilML Workshop, ICML 2026
- Pluralistic Alignment Workshop, ICML 2026
- Post-AGI Science and Society (P-AGI) Workshop, ICLR 2026
- IEEE Canadian Journal of Electrical & Computer Engineering (CJECE)
- ACM Transactions on Intelligent Systems and Technology (TIST)

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML/CSS

ML frameworks: PyTorch, JAX, TensorFlow

Scientific stack: NumPy, Pandas, Matplotlib

Spoken languages: English (fluent), Persian (native), Azeri (native)